

Reversible instance normalization

Executive summary of superseded evidence

Current decision. Do not use the 94 migrated configurations as current-model evidence. They were produced by the superseded flat PatchTST implementation, before the pinned source-adapted package became the sole current backbone. Tables 2 and 3 preserve the previously analyzed signal, but every affected comparison must be rerun before it supports a current conclusion.

Evidence completeness

Archived core evidence	Datasets	Settings	Methods	Seeds	Complete seed runs
Traffic, Electricity, Solar; PatchTST	3	5	6	3	270 / 270

Table 1: The 90 historical method configurations have non-empty metrics, dataset configuration, completion signature, and criterion plot for all three seeds. The 94 schema-1 manifests consist of these 90 configurations plus four isolated test configurations. Schema validity does not make their superseded model implementation scientifically reusable.

Aggregate normalization result

Method	Test1 gain	Test2 gain
Instance normalization, MSE	14.44%	16.50%
Instance normalization, nMSE	17.51%	16.65%
Affine RevIN, MSE	14.57%	16.58%
Affine RevIN, nMSE	17.57%	16.70%
Validation-selected method	16.20%	16.75%
Optimistic test oracle	17.72%	17.05%

Table 2: Mean relative MSE improvement over global standard normalization, equal-weighting all 15 dataset/setting rows. An instance-normalized method is best on 15/15 Test1 rows and 14/15 Test2 rows; global standard wins only Solar 336:720 on Test2.

Affine and training-loss ablations

Paired comparison over 15 settings	Test1 median	Test1 max	Test2 median	Test2 max
affine-non-affine , MSE training	0.097%	0.739%	0.069%	0.455%
affine-non-affine , nMSE training	0.073%	0.239%	0.051%	0.204%
nMSE minus MSE training, non-affine	Test1		Test2	
Mean relative effect	+4.147%		−0.019%	
Median relative effect	+0.545%		+0.314%	
Rows improved	12/15		10/15	
Worst row	−0.855%		−7.833%	

Table 3: The affine layer is numerically negligible. Normalized-loss training has a positive Test1 mean driven by Electricity but no aggregate Test2 gain; its worst Test2 row is Electricity 168:24.

Run-level uncertainty

Example	Seed 1	Seed 2	Seed 3	Mean	Sample SD
Electricity 504:168, Test1 MSE, non-affine instance	11.07M	0.261M	0.221M	3.85M	6.25M

Table 4: The run seed fixes user partition, sampling, initialization, and optimization together. Reported seed variation is therefore total run uncertainty rather than optimizer-only uncertainty.

Evidence retained before deleting artifacts

Retained conclusion	Quantitative basis
Use instance normalization	14.44–17.57% Test1 and 16.50–16.70% Test2 gain versus standard normalization in Table 2.
Do not require affine RevIN	Median paired effects are 0.051–0.097%, with all maxima below 0.74%, in Table 3.
Keep nMSE selectable	+4.147% mean on Test1 but −0.019% on Test2, with a −7.833% worst row, in Table 3.
Report total run uncertainty	The representative three-seed spread is 0.221M–11.07M and SD 6.25M in the uncertainty table.

Historical boundary. These values remain an analyzed record of the pre-package corrected protocol, not evidence from the current pinned PatchTST/DLinear implementation and not a numerical reproduction of the rejected ECML/AALTD paper. The original submission remains read-only guidance in the workspace-level `latex/submissions` directory.